Banknote Recognition

In this project, we have two goals:

1. Classify fake or real banknotes
2. Determine the amount of banknote

Note: Banknote recognition of dollar currency is our priority for now but we might aware that after a short or long time, we need to recognize other currency banknotes with our system. So, we have to prepare an appropriate architecture for future use for different currencies.

We can consider two approaches to achieve our goals in this project.

1. Parallel
   In parallel, we can process both of the goals simultaneously.
2. Sequential
   In sequential, we first detect if the banknote is real or not. If it was real, then we'll continue to determine the amount of the correlated banknote.

Both of the approaches have some advantages and disadvantages. Therefore, we need to move on and implement them and we'll find out which one is better to use according to our needs and expectations in testing and evaluation step.

Fig. 1, 2 both display a simple flowchart of these two approaches.

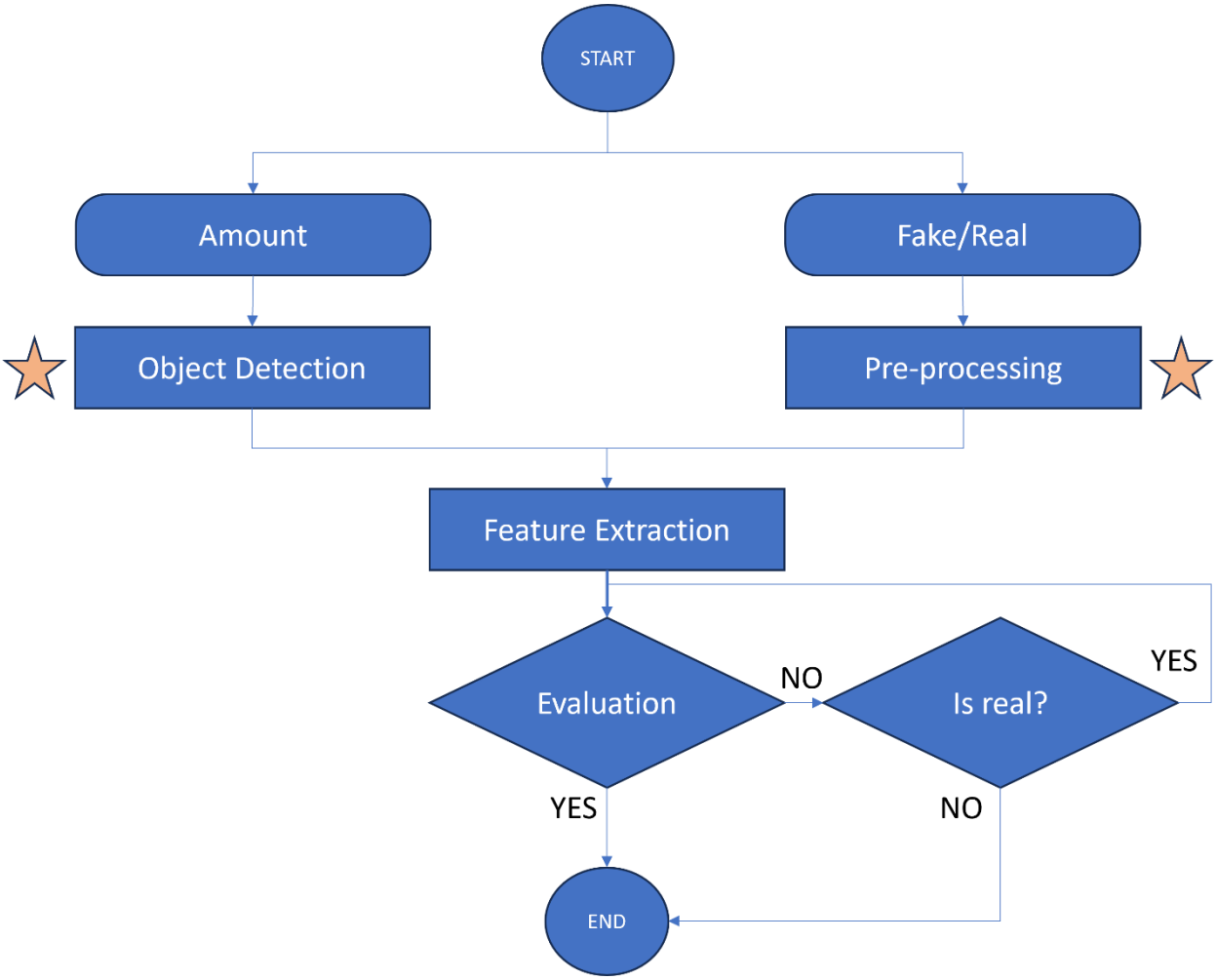


Fig. 1 Flowchart of parallel approach

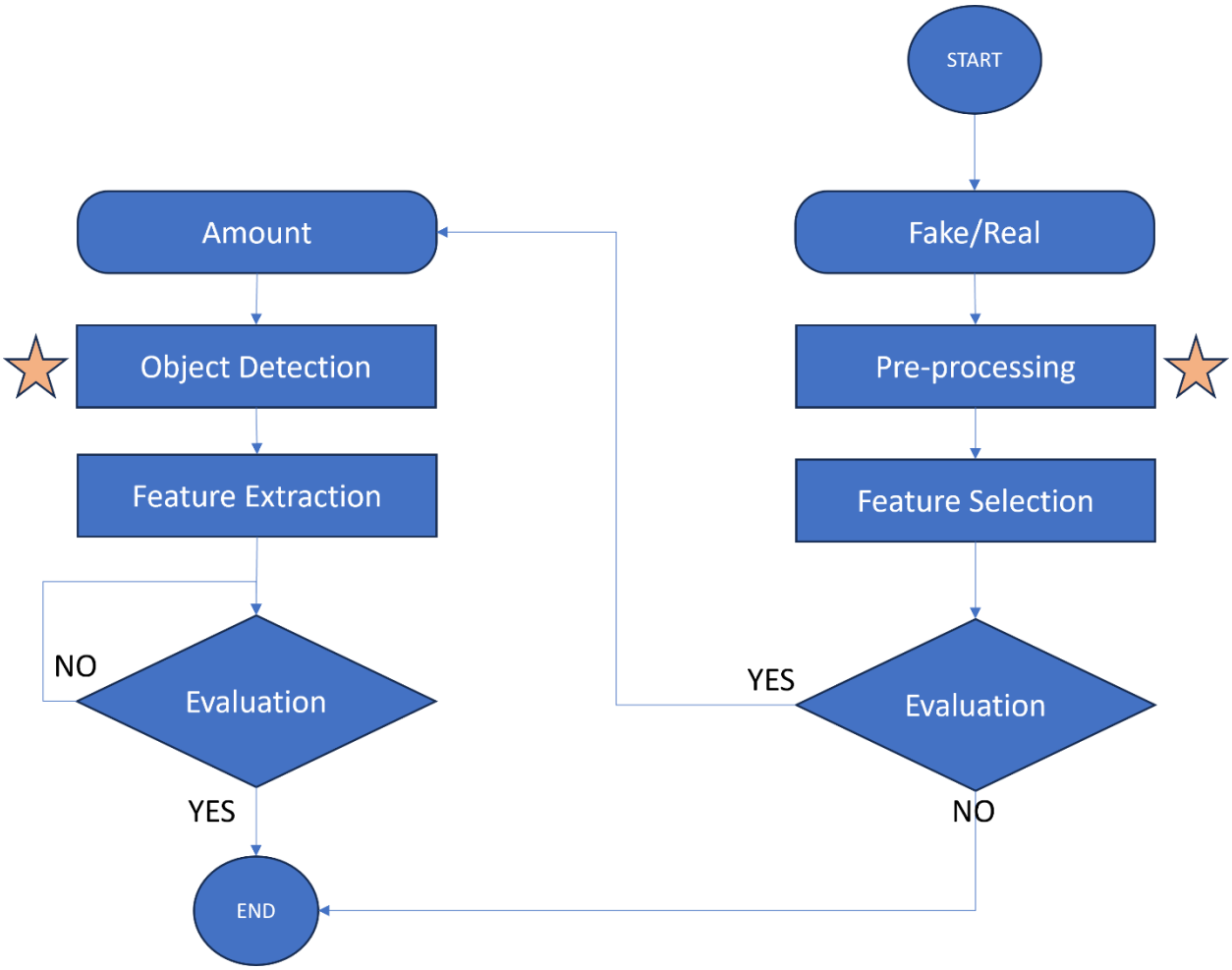


Fig. 2 Flowchart of sequential approach